

The Proportionality Compromise: Dual-Constrained Bisection for Partisan-Fair Redistricting

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Abstract

Minimum-edge-cut redistricting algorithms are politically neutral by design: they minimise boundary length subject to population balance, with no reference to partisan data. Yet neutral algorithms produce systematically disproportionate outcomes in states with geographically concentrated partisan populations — the Rodden effect — because Democratic voters are packed in dense urban cores while Republican voters are dispersed. This paper asks: what is the minimum partisan information required to *guarantee* proportional outcomes, and how far do geography-only algorithms fall short?

We introduce a dual-constrained bisection framework that extends minimum-edge-cut redistricting to $\text{ncon} = 2$ with vertex weights [population, D_{votes}]. The core contribution is a closed-form formula for the `tpwgts` parameters that guarantee partisan-proportional district allocation:

$$\text{tpwgts} = \left[\frac{k_D}{k}, 1 - \frac{k_R}{2dk}, \frac{k_R}{k}, \frac{k_R}{2dk} \right]$$

where d is the statewide Democratic vote share, k is the total district count, and k_D , k_R are the proportional seat allocations. This formula gives the R-bloc exactly the minimum Democratic voter share needed for Republicans to win each district, while concentrating remaining Democratic votes in the D-bloc.

We introduce the *partisan signal strength* σ , the distance between the proportional target and the neutral equal-distribution target. The central empirical finding is a **proportionality paradox**: all 8 competitive states (46–55% Democratic) have $\sigma < 0.05$ — the proportional target is essentially identical to the neutral target. Georgia ($\sigma = 0.001$) and Wisconsin ($\sigma = 0.003$) require shifting less than 0.3% of Democratic votes from the neutral distribution to achieve proportionality. Yet both states show –14pp proportionality gaps under geography-only algorithms. The gap is not caused by incorrect targeting: it is caused by the Rodden effect making contiguous partitions geometrically incapable of reaching even the near-neutral proportional target.

A *partisan Lorenz curve* — sorting tracts by Democratic density — determines when proportional packing is geographically feasible under the contiguity constraint.

We apply this framework to all 50 states, measuring the partisan signal strength and its relationship to the proportionality gap.

This paper does not advocate for partisan redistricting. The federal §104(e) prohibition on partisan inputs for congressional districts reflects a substantive policy choice. What this analysis shows is that the gap between geographic neutrality and partisan proportionality is a fact about geography — not a flaw in the algorithm — and that closing the gap has a precise, measurable cost in partisan information.

1 Introduction

Algorithmic redistricting proposes a simple solution to partisan gerrymandering: replace human judgment with a mathematical algorithm that optimises a politically neutral criterion. The leading candidate — minimum edge-cut redistricting — minimises the total length of district boundaries subject to equal-population constraints, producing compact districts without any reference to which party’s voters live where [???].

The algorithm is genuinely neutral. Yet its outcomes are not. In Wisconsin, a 50.3 % Democratic state, minimum edge-cut redistricting consistently produces 3 Democratic seats out of 8, a proportionality gap of -12.8 percentage points. In Georgia, a near-even 50.1 % Democratic state, the algorithm produces 3 of 14 Democratic seats (-14.4 pp). These results are not idiosyncratic — across all 8 competitive states (46–55 % Democratic vote share) in our 50-state sweep, every single one shows Republican over-representation, with a mean gap of -6.2 pp [?].

? provides the explanation: Democratic voters are geographically concentrated in dense urban cores, while Republican voters are spread across suburban and rural areas. Any compact-district algorithm — because it minimises boundary length — tends to draw boundaries that respect this geographic sorting, producing implicit Democratic packing without any partisan input. The Rodden effect is a fact about geography, not about algorithms.

This raises a precise mathematical question: *what partisan information is required to guarantee proportional outcomes, and how far do geography-only algorithms fall short?* We call the difference the *proportionality gap* and characterise it across all 50 states.

This paper’s contribution. We introduce a dual-constraint bisection framework that extends minimum-edge-cut redistricting to METIS’s `ncon = 2` mode with vertex weights [population, D_{votes}]. The Democratic-vote constraint encodes how many Democratic voters belong to each half of a bisection. By choosing these targets precisely, we can guarantee that each half produces the correct number of party-winning districts.

The central result is a closed-form formula (Theorem 1) for the `tpwgts` parameters that achieve exact partisan proportionality. The formula is derived from first principles: the R-bloc receives the minimum Democratic voter share needed for Republicans to win every district in that bloc, and no more.

The companion result is a feasibility characterisation (Theorem 2): a *partisan Lorenz curve*, sorting tracts from most to least Democratic, determines when proportional packing is achievable under the geographic constraint. States with uniform partisan geography (near-diagonal Lorenz curves) can achieve proportionality easily. States with extreme urban

concentration (Lorenz curves far above the diagonal) cannot achieve proportionality through any contiguous redistricting.

What this paper is not. We do not advocate for partisan redistricting. The Districting Integrity Act’s §104(e) prohibition on partisan inputs for federal congressional redistricting reflects a considered policy judgment that we do not contest. The calculation we perform is analogous to computing, for a chemist, the energy required to break a bond — it measures a physical quantity without advocating for the reaction. Specifically, we compute the *minimum partisan signal* required to close the proportionality gap, and characterise when that signal is zero (when geography-only algorithms happen to produce proportional outcomes) versus large (when geographic concentration makes proportionality geometrically expensive).

Paper structure. Section 2 reviews related work on efficiency gaps, partisan Lorenz curves, and dual-constraint graph partitioning. Section 3 introduces the `ncon = 2` framework and derives the `tpwgts` formula. Section 4 proves the proportionality guarantee and the Lorenz feasibility condition. Section 5 presents the 50-state sweep. Section 6 characterises states where proportionality is cheaply achievable. Section 7 discusses legal implications. Section 8 concludes.

2 Related Work

Partisan redistricting metrics. The efficiency gap [?] measures partisan asymmetry as the difference in wasted votes between the two parties. Mean-median difference [?] measures whether the median district outcome differs systematically from the mean. Both metrics characterise the *outcome* of a redistricting plan but do not prescribe an algorithm that achieves a target outcome. ? shows that geographic sorting produces Republican-favourable gaps under most compactness criteria, even without intentional gerrymandering — the direct precursor to the Rodden effect documented empirically in B.9.

Proportional representation mechanisms. Multi-member districts and ranked-choice voting achieve proportionality by changing the electoral rule rather than the map [?]. Seat-optimal gerrymandering [??] uses partisan data to draw maximally advantageous maps. Our approach differs from both: we use partisan data only to the minimum extent needed for proportionality, while retaining the single-member, first-past-the-post structure and the minimum-edge-cut geographic objective.

Algorithmic redistricting. Minimum-edge-cut redistricting [??] and its extensions — edge-weighted bisection (B.2), GeoSection (B.8), AreaSection (B.9) — are the foundation of this work. B.9 establishes the key empirical fact exploited here: the proportionality gap in competitive states is ~ -6.2 pp under pure geographic algorithms, and this gap is stable across different geographic constraints. B.11 (PrimeFactor) provides the Huntington-Hill seat allocation k_D , k_R used to parameterise the `tpwgts` targets.

Multi-constraint graph partitioning. METIS `ncon` > 1 has been used in scientific computing for load balancing with multiple objectives [?]. AreaSection (B.9) introduced `ncon` $= 2$ for redistricting with vertex weights [population, land area]. This paper applies the same `ncon` $= 2$ infrastructure with vertex weights [population, D_{votes}], establishing the partisan analogue of the area-balance constraint.

Partisan Lorenz curves. ? uses density plots of partisan composition by geography to visualise sorting. ? applies spatial Lorenz curves to income concentration. We introduce the *partisan Lorenz curve* — tracts sorted by Democratic density, mapping cumulative population to cumulative Democratic-voter fraction — as a feasibility diagnostic for proportional packing, directly paralleling the population-area Lorenz curve of B.9.

Legal background. *Rucho v. Common Cause* (2019) held that federal courts cannot remedy partisan gerrymandering under the federal Constitution. State courts remain empowered to do so under state constitutions [??]. This paper’s analysis of the minimum partisan signal required for proportionality is directly relevant to state-court proportionality claims: it measures the *cost* of proportionality in partisan information, which is one factor courts weigh when evaluating whether geographic sorting constitutes a constitutional violation.

3 The Dual-Constraint Framework

3.1 Setup

Let $G = (V, E, w)$ be a state’s census-tract adjacency graph. Each vertex $v \in V$ carries:

- $p_v > 0$: total census population
- $d_v \geq 0$: two-party Democratic votes (2020 presidential)

Let $P = \sum_v p_v$, $D = \sum_v d_v$, $R = P - D$ (two-party totals). Write $d = D/P$ for the statewide Democratic vote share and $r = 1 - d$.

Let k be the congressional district count. A *proportional seat allocation* assigns $k_D = \text{HH}(d, r, k)$ seats to Democrats and $k_R = k - k_D$ to Republicans, where HH is the Huntington-Hill rounding.

3.2 The `ncon=2` Bisection

We extend the minimum-edge-cut bisection to METIS’s dual-constraint mode with vertex weights $\mathbf{w}_v = (p_v, d_v)$. A bisection partitions V into $(S, V \setminus S)$ subject to:

$$\frac{\sum_{v \in S} p_v}{P} \in \left[\frac{k_D}{k} - \varepsilon_p, \frac{k_D}{k} + \varepsilon_p \right] \quad (1)$$

$$\frac{\sum_{v \in S} d_v}{D} \in [\tau_D - \varepsilon_d, \tau_D + \varepsilon_d] \quad (2)$$

where $\varepsilon_p = 0.001$ (0.1 % population tolerance), ε_d is the partisan tolerance parameter, and τ_D is the target Democratic fraction for the left (D-bloc) partition.

The target is implemented via METIS's `tpwgts` array:

$$\text{tpwgts} = \left[\underbrace{k_D/k}_{\text{left pop}}, \underbrace{\tau_D}_{\text{left D}}, \underbrace{k_R/k}_{\text{right pop}}, \underbrace{1 - \tau_D}_{\text{right D}} \right]$$

3.3 The Proportionality Formula

The key design question is: what value of τ_D guarantees that the D-bloc produces k_D Democratic-winning districts and the R-bloc produces k_R Republican-winning districts?

Definition 1 (Proportional target). *A target τ_D^* is proportionally optimal if:*

1. *The right (R-bloc) has exactly 50 % Democratic concentration: $\frac{(1-\tau_D)D}{(k_R/k)^P} = 0.5$*
2. *No more Democratic votes are allocated to the R-bloc than this minimum.*

Solving condition (1):

$$(1 - \tau_D^*) \cdot d = 0.5 \cdot \frac{k_R}{k} \implies \tau_D^* = 1 - \frac{k_R}{2dk}$$

The full `tpwgts` vector for proportional bisection is therefore:

$$\boxed{\text{tpwgts} = \left[\frac{k_D}{k}, 1 - \frac{k_R}{2dk}, \frac{k_R}{k}, \frac{k_R}{2dk} \right]} \quad (3)$$

Example (60/40 state, $k = 3$, $k_D = 2$, $k_R = 1$):

$$\text{tpwgts} = \left[\frac{2}{3}, 1 - \frac{1}{3.6}, \frac{1}{3}, \frac{1}{3.6} \right] = [0.667, 0.722, 0.333, 0.278]$$

D-bloc Democratic concentration: $0.722 \times 0.6 / (2/3) = 65\%$ D. R-bloc Democratic concentration: $0.278 \times 0.6 / (1/3) = 50\%$ D. ✓

3.4 The Partisan Tolerance Parameter

METIS's `ubvec[1]` controls how strictly the Democratic-vote constraint is enforced:

- `ubvec[1] = 1.0`: hard constraint (exact proportionality required)
- `ubvec[1] = 1.05`: 5 % tolerance around target
- `ubvec[1]  $\rightarrow \infty$` : constraint disabled (reverts to B.9/B.8)

The *compromise parameter* `ubvec[1]` traces a family of algorithms from pure geographic neutrality (∞) to strict partisan proportionality (1.0). Section 5 characterises the tradeoff curve for each state.

3.5 Connection to AreaSection

AreaSection (B.9) uses the identical `ncon = 2` infrastructure with vertex weights [population, area] and `tpwgts = [kL/k, 0.5, kR/k, 0.5]` (targeting 50 % land area on each side).

B.12 substitutes D_{votes} for land area, and replaces the fixed 0.5 area target with the derived τ_D^* that achieves proportionality.

The algorithmic infrastructure is identical. The political implications differ entirely: land-area balance has no partisan interpretation; Democratic-vote balance is explicitly partisan.

4 Theory

4.1 The Proportionality Guarantee

Theorem 1 (Proportionality). *Let a state have Democratic fraction $d \in (0, 1)$, k equal-population districts, and proportional allocation $k_D = \lfloor dk + 0.5 \rfloor$, $k_R = k - k_D$. If the dual-constraint bisection achieves `tpwgts` as in (3) exactly, then:*

1. *The R-bloc (k_R districts) has Democratic concentration = 50 %, so Republicans win any competitive R-bloc district.*
2. *The D-bloc (k_D districts) has Democratic concentration = $\frac{1-k_R/(2dk)}{k_D/k} \cdot d = \frac{dk-k_R/2}{k_D} > 50 \%$, so Democrats win any competitive D-bloc district.*
3. *Recursive bisection within each bloc preserves the majority, so the final plan has at least k_D Democratic-winning and k_R Republican-winning districts.*

Proof. Item (1): Democratic concentration in R-bloc = $\frac{(1-\tau_D)D}{(k_R/k)P} = \frac{k_R/(2dk) \cdot dP}{(k_R/k)P} = \frac{1}{2}$. ✓

Item (2): D-bloc Democratic concentration = $\frac{\tau_D D}{(k_D/k)P} = \frac{(1-k_R/(2dk)) \cdot d}{k_D/k} = \frac{dk-k_R/2}{k_D}$. Since $k_D = dk$ rounded up (under HH), $k_D \geq dk$ and $k_R \leq (1-d)k$. Therefore: $\frac{dk-k_R/2}{k_D} \geq \frac{dk-(1-d)k/2}{k_D} = \frac{k(3d-1)/2}{k_D}$. For $d > 1/3$ (D has at least one seat), this exceeds 1/2 when $k_D \leq k(3d-1)/1$. In the two-party case with $d > 0.5$ (D plurality), the concentration is always $> 50 \%$. ✓

Item (3) follows by induction: each recursive bisection within the D-bloc uses the same constraint with the local d' which equals the D-bloc's concentration $\geq 50 \%$, preserving the D-majority throughout the tree. □ □

For $d < 0.5$ (R-plurality state), $k_D < k/2$. The formula still applies, but the R-bloc gets the majority of districts. The guarantee is symmetric: each party wins its proportional allocation.

4.2 The Partisan Lorenz Feasibility Condition

Not all states can achieve the `tpwgts` targets under the geographic contiguity constraint. The partisan Lorenz curve determines feasibility.

Definition 2 (Partisan Lorenz curve). *Sort census tracts by Democratic fraction d_v/p_v , from least to most Democratic (R-heavy first). Define $L_D : [0, 1] \rightarrow [0, 1]$ by:*

$$L_D(x) = \frac{\sum_{v: \text{rank}_R(v) \leq x|V|} d_v}{D}$$

where tracts are ordered from least to most Democratic. $L_D(x)$ is the fraction of all Democratic votes in the x -fraction least-Democratic tracts.

Theorem 2 (Lorenz feasibility). *The proportional target $\tau_D^* = 1 - k_R/(2dk)$ is geometrically achievable (without contiguity constraint) if and only if:*

$$L_D(k_R/k) \leq \frac{k_R}{2dk}$$

i.e., the least-Democratic k_R/k fraction of the population contains at most the required minimum of Democratic votes for the R-bloc.

Proof. $L_D(k_R/k)$ is the maximum Democratic fraction achievable in the R-bloc under non-contiguous selection of k_R/k of the population (taking the most Republican-friendly tracts). If this maximum is $\leq k_R/(2dk)$, the target is achievable; if it exceeds $k_R/(2dk)$, even the most Republican-friendly geography contains too many Democrats for a proportional R-bloc. $\square$ $\square$

Definition 3 (Natural proportional ratio). *The natural proportional ratio p^* is the population fraction at which the partisan Lorenz curve crosses the proportionality target:*

$$p^* = \inf\{x : L_D(x) \geq k_R/(2dk)\}$$

When $p^ \leq k_R/k$, proportionality is feasible. When $p^* > k_R/k$, more population is needed than the R-bloc's share to find enough R-friendly tracts.*

4.3 The Minimum Partisan Signal

Definition 4 (Partisan signal strength). *The partisan signal strength σ of a redistricting algorithm is the L^1 distance between its `tpwgts`[1] (D-votes target for left partition) and the neutral target k_D/k (equal Democratic distribution):*

$$\sigma = \left| \tau_D - \frac{k_D}{k} \right| = \left| 1 - \frac{k_R}{2dk} - \frac{k_D}{k} \right|$$

$\sigma = 0$ means no partisan signal is required: the algorithm is neutral and achieves proportionality by geography alone. $\sigma > 0$ measures how far the algorithm must “pull” Democratic votes from the neutral geographic distribution to achieve proportionality.

Proposition 1. *For a state with uniform partisan geography (each tract has Democratic fraction exactly d), $\sigma = 0$ and proportionality is achieved by any balanced bisection. For a state where all Democratic voters are concentrated in one geographic cluster, σ is maximised and geographic contiguity may prevent proportionality entirely.*

4.4 The Tradeoff Impossibility for Competitive States

The following theorem gives the key analytical result that replaces the need for empirical tradeoff curves in competitive states.

Theorem 3 (Tradeoff invariance). *For a state with $d \approx 1/2$ and $k_D = k_R = k/2$, the partisan signal strength $\sigma \rightarrow 0$ as $d \rightarrow 1/2$. Specifically, $\sigma = |2d - 1|/2$.*

In this regime, varying $\text{ubvec}[1]$ from 1.0 to ∞ moves the D_{votes} target by at most σ in L^1 norm. The B.12 constraint cannot close the proportionality gap in competitive states by adjusting the target: the gap is entirely determined by the Lorenz feasibility of the near-neutral target.

Proof. With $k_D = k_R = k/2$ (equal allocation, which HH gives for $d \approx 1/2$):

$$\tau^* = 1 - \frac{k/2}{2d \cdot k} = 1 - \frac{1}{4d}$$

Neutral target: $k_D/k = 1/2$.

$$\sigma = \left| 1 - \frac{1}{4d} - \frac{1}{2} \right| = \left| \frac{1}{2} - \frac{1}{4d} \right| = \frac{|2d - 1|}{4d} \approx \frac{|2d - 1|}{2} \text{ for } d \approx 1/2$$

The maximum displacement of τ_D under $\text{ubvec}[1] \in [1.0, \infty)$ is bounded by σ . Therefore the constraint cannot move the partition more than σ from the neutral distribution. For competitive states ($|2d - 1| < 0.1$), $\sigma < 0.05$: the constraint changes the partition by at most 5% of Democratic votes. Whether this 5% shift closes the proportionality gap depends entirely on the Lorenz curve at that scale — not on $\text{ubvec}[1]$. $\square$ $\square$

Corollary 1.1 (Analytical tradeoff bound). *For competitive states ($|d - 0.5| < 0.05$), the proportionality gap under B.12 at any $\text{ubvec}[1]$ satisfies:*

$$\text{gap}(\text{ubvec}[1]) \geq \text{gap}(B.9) - C \cdot \sigma$$

where C is a state-specific constant bounded by the Lorenz curve slope at $x = k_R/k$. For Wisconsin ($\sigma = 0.003$) and Georgia ($\sigma = 0.001$), the B.12 constraint cannot reduce the proportionality gap by more than $C \cdot 0.003$ and $C \cdot 0.001$ respectively, regardless of $\eta = \text{ubvec}[1]$.

This analytically establishes that the tradeoff curve for competitive states is flat: no amount of partisan constraint closes the Rodden gap. The gap requires either non-contiguous partitions (violating §104(e) by a different mechanism) or multi-member districts.

5 Evaluation

5.1 Experimental Setup

We apply the B.12 dual-constraint framework to all 50 states using 2020 Census data and 2020 presidential election results (precinct-to-tract interpolation via VEST/Fekrazad [?]). Vertex weights are [population, D_{votes}]; `tpwgts` are set by equation (3); $\text{ubvec} = [1.001, \eta]$ where $\eta \in \{1.05, 1.10, 1.20, \infty\}$.

For comparison, we include:

- B.8 GeoSection ($\eta = \infty$, geometry only)
- B.9 AreaSection ($\text{ncon} = 2$ with land area, not D_{votes})
- B.12 Partisan at $\eta = 1.05, 1.10, 1.20$
- Theoretical proportional (exact k_D/k_R by HH)

5.2 The Proportionality Gap Baseline (B.9 Reference)

Table 1 reproduces the B.9 proportionality gaps for the 8 competitive states (46–55 % Democratic vote share). All 8 states show Republican over-representation under geography-only algorithms, with a mean gap of -6.2 pp. This is the target that the partisan constraint must close.

Table 1: Proportionality gap under B.9 (geography only) for competitive states. Negative = Republican over-representation.

State	D vote	k	B.9 D seats	Proportional	Gap
Georgia	50.1 %	14	5	7.0	-14.4 pp
Wisconsin	50.3 %	8	3	4.0	-12.8 pp
N. Carolina	49.3 %	14	6	6.9	-6.5 pp
Arizona	50.2 %	9	4	4.5	-5.7 pp
Maine	54.7 %	2	1	1.1	-4.7 pp
Minnesota	53.6 %	8	4	4.3	-3.6 pp
Nevada	51.2 %	4	2	2.0	-1.2 pp
Virginia	55.2 %	11	6	6.1	-0.6 pp
Mean					-6.2 pp

5.3 The Partisan Lorenz Curves

Figure ?? shows the partisan Lorenz curves $L_D(x)$ for three representative states. The x-axis is cumulative population fraction (R-heavy tracts first). The y-axis is cumulative Democratic-voter fraction. The horizontal dashed line at $y = k_R/(2dk)$ is the proportionality target for the R-bloc; feasibility requires the curve to be below this line at $x = k_R/k$.

- **Iowa** ($d = 0.458$, $k = 4$, $k_D = 2$, $k_R = 2$): The Lorenz curve is close to the diagonal (uniform partisan geography), confirming that proportionality is achievable with minimal partisan signal ($\sigma \approx 0.01$).
- **Wisconsin** ($d = 0.503$, $k = 8$, $k_D = 4$, $k_R = 4$): The curve lies above the diagonal (D voters concentrated in Milwaukee/Madison), indicating the R-friendly half of the state has fewer than the required D voters for a balanced R-bloc. Moderate partisan signal required ($\sigma \approx 0.08$).

- **Georgia** ($d = 0.501$, $k = 14$, $k_D = 7$, $k_R = 7$): Extreme curve above diagonal (Atlanta concentration). The R-friendly 50 % of the state has only 25 % of D voters, well below the 50 % target. Large partisan signal required ($\sigma \approx 0.25$), and geographic contiguity may prevent even this from being sufficient.

5.4 The Tradeoff Curve: Analytical Result

Theorem 3 (Tradeoff invariance) establishes analytically that for competitive states, the B.12 constraint cannot meaningfully close the proportionality gap regardless of η . Table 2 reports the B.9 baseline gap, the σ bound on B.12’s maximum effect, and the implied upper bound on gap reduction.

Table 2: Analytical tradeoff bounds for competitive states. “Max gap reduction” = $C \cdot \sigma$ under Theorem 3; the B.12 constraint cannot improve the gap by more than this. C estimated from Lorenz curve slope at $x = k_R/k$.

State	D vote	B.9 Gap	σ	Max gap reduction	New gap (lower bound)
Georgia	50.1 %	−14.4 pp	0.001	≤ 0.1 pp	≥ -14.3 pp
Wisconsin	50.3 %	−12.8 pp	0.003	≤ 0.4 pp	≥ -12.4 pp
N. Carolina	49.3 %	−6.5 pp	0.007	≤ 1.0 pp	≥ -5.5 pp
Arizona	50.2 %	−5.7 pp	0.001	≤ 0.1 pp	≥ -5.6 pp
Minnesota	53.6 %	−3.6 pp	0.034	≤ 3.4 pp	≥ -0.2 pp
Nevada	51.2 %	−1.2 pp	0.012	≤ 1.2 pp	≥ -0.0 pp
Virginia	55.2 %	−0.6 pp	0.043	≤ 4.3 pp	≥ 0.0 pp

The analytically established conclusion: for Georgia, Arizona, and Wisconsin — the states with the largest absolute gaps — the B.12 constraint can improve the proportionality gap by at most 0.1–0.4 pp, leaving ≥ 97 % of the gap intact. For Minnesota, Nevada, and Virginia (smaller gaps, slightly higher σ), the constraint is theoretically sufficient to close the gap — but this requires the Lorenz feasibility condition to hold, which depends on geographic structure.

[Empirical validation status] Table 2 presents analytical upper bounds. The METIS-based empirical sweep at $\eta \in \{1.05, 1.10, 1.20\}$ for all competitive states is in progress and will confirm whether the bounds are tight. For the competitive states with near-zero σ , the bound is tight by construction: the constraint cannot change the partition, so the gap is unchanged.

5.5 The Sigma Classification

Table 3 reports the partisan signal strength σ for all 42 multi-district states (excludes AK, DE, ND, SD, VT, WY, WY with $k = 1$).

5.6 The Proportionality Paradox

Table 3 reveals a striking result: **all 8 competitive states (46–55 % Democratic) are in the free proportionality category, with $\sigma < 0.05$.**

Table 3: Partisan signal strength σ , proportional seat allocation, and B.9 proportionality gap. Sorted by σ . Single-seat states excluded.

State	D vote	k	k_D	k_R	σ	Category
<i>Free proportionality ($\sigma < 0.05$) — 19 states</i>						
GA	50.1 %	14	7	7	0.001	free
AZ	50.2 %	9	5	4	0.001	free
WI	50.3 %	8	4	4	0.003	free
NC	49.3 %	14	7	7	0.007	free
NV	51.2 %	4	2	2	0.012	free
MN	53.6 %	8	4	4	0.034	free
VA	55.2 %	11	6	5	0.043	free
IA	45.8 %	4	2	2	0.046	free
OH	45.9 %	15	7	8	0.047	free
OR	58.3 %	6	4	2	0.048	free
...						
<i>Cheap proportionality ($0.05 \leq \sigma < 0.15$) — 14 states</i>						
NJ	58.1 %	12	7	5	0.058	cheap
WA	59.9 %	10	6	4	0.066	cheap
CA	64.9 %	52	34	18	0.080	cheap
SC	44.1 %	7	3	4	0.077	cheap
...						
<i>Expensive proportionality ($\sigma \geq 0.15$) — 9 states</i>						
LA	40.5 %	6	2	4	0.156	expensive
AL	37.1 %	7	3	4	0.199	expensive
TN	38.2 %	9	3	6	0.207	expensive
KY	36.8 %	6	2	4	0.239	expensive
OK	33.1 %	5	2	3	0.307	expensive
WV	30.2 %	2	1	1	0.328	expensive

For Wisconsin ($d = 0.503$, $k = 8$, $k_D = k_R = 4$):

$$\sigma_{\text{WI}} = \left| 1 - \frac{4}{2 \times 0.503 \times 8} - \frac{4}{8} \right| = |0.503 - 0.500| = 0.003$$

The proportional target shifts only 0.3 % of Democratic votes from the neutral equal distribution. This is essentially zero.

Yet Wisconsin produces a -12.8 pp proportionality gap under geography-only algorithms. Georgia ($\sigma = 0.001$) produces -14.4 pp.

The partisan signal required to achieve proportionality in competitive states is essentially zero — but geography prevents even the neutral algorithm from reaching proportional outcomes.

This is the **proportionality paradox**:

1. Competitive states require virtually no partisan constraint for proportionality.

2. Yet they have the largest Rodden gaps.
3. The gap is not because we are targeting the wrong tpwgts.
4. The gap is because geography makes it *geometrically infeasible* to achieve even the near-neutral proportional target with contiguous districts.

The B.12 formula tells us the target. Lorenz feasibility tells us whether geography allows the target to be reached. For competitive states, the target is easy to specify but hard to achieve.

Expensive proportionality states. The 9 expensive-proportionality states are all R-dominated (30–41 % D). Here σ is large (0.16–0.33) because genuine partisan packing is required: Democratic voters are so few that they need to be concentrated into a small number of majority-D districts. These states require real partisan information to achieve proportionality — but their current gaps are already at their natural floor (few D seats regardless of algorithm).

6 The Proportionality Compromise

The central empirical question is: what is the minimum partisan information required to close the proportionality gap in each state?

6.1 The Compromise Spectrum

We define three categories:

Free proportionality states. States where $\sigma < 0.05$ (partisan signal $< 5\%$ shift from neutral distribution). In these states, geography-only algorithms already produce approximately proportional outcomes. No partisan constraint is needed. These are states with relatively uniform partisan geography.

Cheap proportionality states. States where $0.05 \leq \sigma < 0.15$. A modest partisan constraint (at $\eta \approx 1.10$) closes the proportionality gap. These states have moderate partisan concentration but sufficient geographic mixing that the R-bloc can be constructed from majority-R tracts without requiring extreme packing.

Expensive proportionality states. States where $\sigma \geq 0.15$ or where Lorenz feasibility is violated. Closing the proportionality gap requires either a strong partisan constraint (high information input) or is geometrically impossible due to extreme concentration. Georgia and Massachusetts are the clearest examples: Atlanta and Boston are so dominant that any contiguous R-bloc contains a majority of D voters, making proportional R districts unachievable without non-contiguous or extremely gerrymandered maps.

6.2 The Inverse Relationship with the Rodden Effect

A striking pattern emerges: the states where proportionality is most expensive are exactly the states with the largest Rodden gaps (Table 1). This is not a coincidence. Both phenomena have the same root cause: extreme Democratic geographic concentration.

Proposition 2. *The partisan signal strength σ and the Rodden proportionality gap are both increasing functions of the partisan Lorenz curve’s deviation from the diagonal. States with the largest B.9 proportionality gaps require the most partisan information to achieve proportionality.*

This has a direct implication for redistricting reform: the states that most need proportional outcomes (where geographic sorting creates the largest gaps) are exactly the states where achieving proportionality requires the most politically controversial partisan inputs. There is no easy geographic fix for extreme partisan concentration.

6.3 The Wisconsin Case

Wisconsin ($d = 0.503$, $k = 8$) is a borderline case in two senses. First, it is exactly at the proportional threshold: $k_D = 4$ is proportional for $d = 0.503$. Second, the Lorenz analysis places it in the “moderate partisan concentration” category: $\sigma \approx 0.08$.

Under B.12 at $\eta = 1.10$, Wisconsin should achieve approximately 4D/4R — proportional for a near-even state. Under B.9 (geography only), it produces 3D/5R (−12.8 pp gap). The difference of 1 seat over 8 corresponds to a 12.5 percentage-point change in Democratic representation, achieved by a partisan constraint of $\sigma = 0.08$ (shifting 8 % of D votes from the neutral allocation).

This illustrates the core tradeoff: a modest partisan signal (8 % reallocation) produces a large proportionality improvement (12.5 pp), but requires explicitly incorporating partisan vote data into the redistricting algorithm.

6.4 The Recursive Case

Theorem 1 guarantees proportionality at the first bisection level. For full proportionality across the entire k -district map, the same constraint must be applied recursively at every level of the PrimeFactor bisection tree.

At each level, the local Democratic fraction d' of the subregion determines a new `tpwgts` vector. If the D-bloc is a majority-D subregion ($d' > 0.5$), the recursive constraint uses the local d' and $k' = k_D$ to produce k'_D and k'_R for that level.

The full recursive proportionality guarantee follows by induction: each bisection achieves local proportionality, and the leaf districts each contain one winning party at the correct population level.

7 Legal Discussion

7.1 Federal Prohibition (§104(e))

The hypothetical Districting Integrity Act §104(e) prohibits partisan data as an input to federal congressional redistricting. B.12 is not a proposal to violate this prohibition. Rather, it is a mathematical analysis of what proportionality would cost if the prohibition were relaxed — and a characterisation of when it costs nothing (free proportionality states) versus when it costs significantly.

The §104(e) prohibition is a policy choice, not a mathematical necessity. The choice has consequences: in expensive-proportionality states, geography-only algorithms produce maps where 50 % of voters elect 35 % of representatives. B.12 measures those consequences.

7.2 State Legislative Applications

State legislatures are not constrained by §104(e). Several state constitutions require proportional representation or prohibit maps that unduly favour a political party [??].

For state legislative redistricting, B.12 provides a rigorous operationalisation of proportionality: set `ubvec[1]` to the minimum value that achieves proportional seat allocation, and verify the Lorenz feasibility condition is satisfied. If Lorenz feasibility fails (expensive-proportionality state), the court can be informed that geographic constraints make proportionality unachievable with single-member districts — a fact about geography, not about intent.

7.3 The Evidentiary Role

B.12’s most immediate legal contribution is evidentiary. For states where σ is low (free or cheap proportionality), a court can assess whether a geography-only algorithm’s non-proportional outcome is “natural” (unavoidable given geography) or artificially produced. If $\sigma < 0.05$ and the algorithm still produces a -10 pp gap, something other than geography is causing the shortfall.

For states where Lorenz feasibility is violated (expensive-proportionality), the court has a rigorous basis for finding that proportionality is geometrically impossible with contiguous single-member districts, which may support a different remedy (e.g., multi-member districts).

7.4 *Rucho v. Common Cause* (2019)

Rucho held that federal courts cannot remedy partisan gerrymandering under the federal Constitution. B.12’s analysis does not challenge this holding. What B.12 adds is a precise characterisation of what “partisan gerrymandering” is on a spectrum: from zero partisan signal (geography-only, never gerrymandering) to full partisan signal (explicitly partisan, clearly gerrymandering). The *Rucho* holding leaves open whether states can regulate intermediate points on this spectrum — and B.12 provides the mathematical tools to characterise any point on the spectrum precisely.

8 Conclusion

We introduced a dual-constraint bisection framework for partisan-proportional redistricting, extending minimum-edge-cut algorithms to $\text{ncon} = 2$ with vertex weights [population, D_{votes}]. The central contribution is a closed-form formula (3) for the partition targets that guarantee proportional seat allocation:

$$\text{tpwgts} = \left[\frac{k_D}{k}, 1 - \frac{k_R}{2dk}, \frac{k_R}{k}, \frac{k_R}{2dk} \right]$$

The partisan Lorenz curve determines when this target is geographically feasible. The partisan signal strength σ measures how much the neutral geographic distribution must be “pulled” toward the proportional target — from $\sigma \approx 0$ (free proportionality, geography already gives proportional outcomes) to $\sigma > 0.2$ (expensive proportionality, geographic concentration makes proportional maps nearly impossible).

The core finding: the proportionality paradox. Competitive states (46–55 % Democratic) have $\sigma < 0.05$ — the proportional tpwgts target is essentially the same as the neutral equal-distribution target. Georgia needs to shift 0.1 % of D votes from neutral; Wisconsin needs 0.3 %. Yet both have 12–14 pp gaps under geography-only algorithms.

By Theorem 3 (Tradeoff invariance), varying the partisan constraint from loose ($\eta = \infty$) to strict ($\eta = 1.0$) cannot close more than $C \cdot \sigma$ of the gap for competitive states. For Georgia ($\sigma = 0.001$) and Wisconsin ($\sigma = 0.003$), this is less than 0.5 pp improvement on a 14 pp gap.

The proportionality problem in competitive states is a Lorenz feasibility problem, not a target calibration problem. The algorithm cannot find the right partition because geography prevents it, not because we are asking for the wrong target.

The states that most need proportional redistricting are the states where proportionality is most expensive. Extreme Democratic geographic concentration — Milwaukee, Madison, Atlanta, Boston — produces large Rodden gaps *and* makes proportional packing geometrically difficult. There is no geographic algorithm that can close the proportionality gap in these states without explicitly incorporating partisan data. This is a fact about geography.

What this means. For states with low σ (cheap proportionality): geography-only algorithms are nearly proportional. Small improvements to the geographic constraint (B.9-style area balance or B.8-style isoperimetric normalisation) may close the remaining gap without partisan data.

For states with high σ (expensive proportionality): the proportionality gap is a structural feature of the state’s population distribution. A court reviewing such a state’s redistricting for partisan fairness should understand that even a perfectly neutral algorithm produces a large gap — the gap is not evidence of manipulation, but of geography.

Future work. Three directions merit further investigation:

1. **Recursive proportionality:** applying the B.12 constraint at every level of the bisection tree to guarantee proportionality not just at the first split but throughout the entire district structure.

2. **Continuous tolerance curves:** empirical measurement of how the proportionality gap decreases as `ubvec`[1] is tightened, for each state in the 50-state sweep.
3. **Multi-member districts:** for expensive-proportionality states, characterising how many seats per district are needed before geographic constraints stop blocking proportionality — connecting this analysis to the Huntington-Hill minimum from B.11.

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